AI6125-MAS GROUP 7 PROJECT REPORT *

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ABSTRACT

This report presents our group project for AI6125 Multi-Agent Systems, in which we design and evaluate a team of six agents for the Tileworld environment. Tileworld is a dynamic grid-based domain that contains agents, tiles, holes, obstacles, and fuel stations, where agents must collect and deliver tiles to holes under sensing, movement, and fuel constraints with a maximum tile carriage of 3. Our work investigates how different agent roles, planning strategies, and communication mechanisms can boost the collective performance in an in-accessible (partially observable) and dynamic (time-varying) environment [1]. We describe the rationale for the design of our six-agent system, the collaboration protocol between agents, the experimental settings used for the evaluation, and the resulting performance analysis across multiple Tileworld configurations.

1 Introduction & Background

Multi-agent systems are concerned with how multiple autonomous agents perceive, decide, act, and coordinate in a shared environment. In this project, we study these issues in Tileworld, a classical testbed for evaluating intelligent agents in dynamic environments. According to the project specification[2], Tileworld is a grid environment populated by agents, tiles, obstacles, holes, and fuel stations. Agents can move one cell at a unit of time in the four cardinal directions, carry up to three tiles, and gain reward by moving tiles into hole cells so that the corresponding holes are filled. In addition, every movement consumes fuel, and each agent follows a Sense->Communicate->Plan->Act cycle at every step. These constraints make the environment suitable for studying bounded rationality, cooperation, and task allocation in a team setting.

Tileworld has long been used as a benchmark environment for agent research because it is simple to define, yet rich enough to capture important properties of dynamic decision making. The original Tileworld literature [3] describes it as a highly parameterized and unpredictable environment in which objects may appear and disappear over time, allowing researchers to study how different reasoning or meta-reasoning strategies perform under varying environmental conditions[4]. A key idea is that the effectiveness of an agent system design depends not only on the agent itself, but also on environmental properties such as dynamism, obstacle density, and task difficulty.

For this course project, the provided simulator is implemented in MASON, which is a Java-based agent-based modeling toolkit. The project requires a team of six students, where each member designs one agent and all six agents operate together in the same Tileworld environment. Each agent has a local perception field within a radius defined by

$$\max(|x - X|, |y - Y|) \leq 3,$$

a limited set of actions

$$\{\texttt{WAIT}, \texttt{MOVE_UP}, \texttt{MOVE_DOWN}, \texttt{MOVE_LEFT}, \texttt{MOVE_RIGHT}, \texttt{PICK_UP}, \texttt{DROP}, \texttt{REFUEL}\},$$

* *Contributions*: **Authors are listed alphabetically, all authors contributed to this project equally.**

a carrying capacity of at most three tiles, and optional communication through broadcast messages [2]. These constraints naturally introduce challenges in partial observability, decentralized coordination, memory layer design, and resource management.

Motivated by these properties, our project focuses on designing a six-agent cooperative team that can divide work effectively, share useful local information, reduce redundant exploration, and adapt to different environment configurations. In the following sections, we present our experimental setup, the individual design of each of the six agents, the communication and collaboration mechanisms used within the team, and the corresponding results and analysis.

2 Environment & Experiment Setup

Our experiment is conducted in a tileworld simulator implemented in Java 1.8 with the Java3D v1.5.1 and MASON 14 package. One simulation will run for 5000 steps and we take the final reward as the metric of the agents system; different simulations will start with a randomly generated seed. The reward is defined by the number of holes filled. The initial fuel is fixed at the maximum fuel carriage of 500.

There are 3 different configurations that affect world size, object lifetime, and object creation distribution.
- Parameter Set 1: World Size = 50×50, Object Creation Rate $\sim\mathcal{N}(\mu = 0.2, \sigma = 0.05)$, Lifetime = 100.
- Parameter Set 2: World Size = 80×80, Object Creation Rate $\sim\mathcal{N}(\mu = 2, \sigma = 0.5)$, Lifetime = 30.
- Parameter Set 3: World Size = 120×120, Object Creation Rate $\sim\mathcal{N}(\mu = 5, \sigma = 0.5)$, Lifetime = 50.

3 Agent Design

3.1 Group7AgentBase - Abstract Base Class

We centralized basic Agent behaviors to this base agent class that extends the `TWAgent` class.

Group7AgentBase is a reusable control and coordination layer for all Group 7 agents. It standardizes low-level mobility with A* path planning, path reuse/clearing, obstacle-aware stepping, and a deterministic zig-zag fallback for exploration when no high-level target is available. It also provides fuel-station handling so subclasses can focus on policy rather than survival plumbing. On communication, the base defines the G7P2 protocol and parser, supports directed and broadcast messages, and auto-emits basic messages like discovery and sensor snapshots.

3.2 AgentHanny - The Proactive Learner

This Agent is designed and implemented by Zhang Yanhai, as an agent that dynamically modeling the current environment with observation data gathered from the team and utilizes the estimation in calculating the expected utility in the planning phase.

Planning This agent decides its action based on **utility over plan**, where a plan is an action sequence from the current position. To determine this, we estimate object survival probability and aggregate utilities over all targets in the sequence. Survival analysis [5] was used for estimating the object existing probability and the full derivation is included in the appendix.

AgentHanny will form plans from memory with picking up 1...available-carry-space tiles and filling them to 1...expected-carried-tiles holes, then calculate the expected utility (details in the appendix). Then the plan with highest EU will be selected and executed, with Target Lock send to teammates. The agent action flow can be summarized as follows:

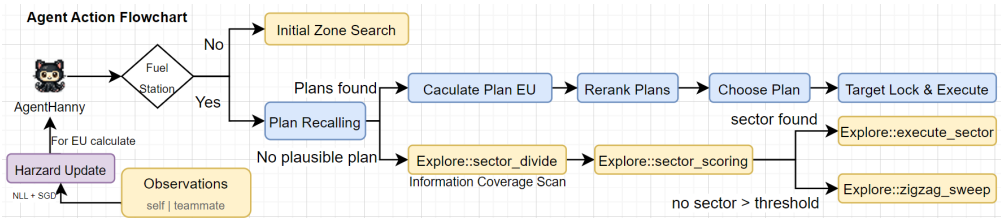

Figure 1: AgentHanny Action Flow

Communications Agent Hanny follows the protocol defined in `Group7AgentBase` and consumes team messages each think cycle. It uses: Target lock is acquired/kept for the *current execution target* (not for every recalled target at plan creation time), renewed before expiry, and released when target is completed, invalidated, pre-empted, or when mode switches (e.g., refuel/re-plan).

Memory Layer In order to make the online MLE possible, AgentHanny keeps a memory side-card to record additional information that `TWAgentWorkingMemory` does not provide. Additional information includes objectType, firstSeen, lastSeen, etc. This side card is always synchronized with the original working memory, and the information is updated when new observation is made.

Exploration In order to maximize the social welfare, AgentHanny is designed to explore the sections of the tileworld map that are least explored by the team. To be more specific, AgentHanny divides the map into a dynamic $n \times n$ grid, where n is set to $WorldSize//15$. Each section is scored with negative signals from teammate density, explored coverage, and distance from current position. Sections whose center has a too-long distance from AgentHanny is excluded. The section with the best score is selected, and if no valid section exists, the agent falls back to zig-zag exploration.

3.3 Agent 2 - <Role / Name>

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3.4 Agent 3 - <Role / Name>

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3.5 Agent 4 - <Role / Name>

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3.6 Agent 5 - <Role / Name>

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3.7 Agent 6 - <Role / Name>

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4 Agent Communication & Collaboration

Our team communication is implemented as a shared protocol layer in `Group7AgentBase` (version tag: `G7P2`), with message fields $\langle version, step, from, to, type, x, y, payload \rangle$. By scheduler design provided by `TWEnvironment`, each agent executes `sense()` and `communicate()` before `think()` and `act()` in every step, so all messages emitted in the communication phase are available to all teammates in the same decision cycle.

The broadcast channel combines lightweight event messages and dense local-state summaries. Event messages include newly observed objects (`NT/NH/OB`), one-time fuel-station discovery (`FS`), and action confirmations (`PK/FH`). In parallel, each agent sends a per-step sensor snapshot (`SS`) that encodes local empty and occupied cells. This hybrid design balances bandwidth and freshness: sparse events reduce redundancy, while `SS` supports rapid correction of stale beliefs.

Collaboration occurs in two stages. During bootstrap, optional zone coordination (`ZA/ZK`) assigns initial search regions via deterministic manager election and bounded retransmission for unacknowledged assignments. After bootstrap, conflict avoidance is handled by target locking (`TL/TR`) with priority and TTL semantics: a cell-level lock is renewed near expiry, released explicitly on completion/cancellation, and removed by timeout if no renewal is observed. This avoids persistent deadlock while reducing duplicate pursuit under partial observability, forming a automatic expanding area of exploration around Fuel Station since Agents will go explore if all targets are consumed or locked.

5 Result and Analysis

The following table summarizes the mean rewards achieved by our full system across the three parameter sets, averaged over 10 runs each:

Design Ablation	Parameter 1	Parameter 2	Parameter 3
G7 Agent System	1086.5	1397.2	2440.8

Table 1: Mean Rewards Achieved Across Different Environment Settings

The increasing absolute reward trend from Parameter Set 1 to 3 is consistent with higher object generation intensity, which creates more service opportunities despite shorter object lifetimes and larger map scale.

The strongest gains arise from coupling memory-based planning with explicit coordination. In the implemented codebase, agents combine local sensing, protocol-level sharing (`SS` and event messages), and role-specific policies built on a common base layer. This integration improves both *throughput* (more successful tile-to-hole completions) and *efficiency* (fewer redundant chases of already-claimed or expired targets). The effect is most visible in dynamic settings, where stale private memory alone is insufficient.

At the same time, performance remains bounded by two practical trade-offs. First, richer reasoning (plan enumeration, lock arbitration, and message processing) introduces additional per-step computation, which can reduce simulation speed on larger maps. Second, communication is step-synchronous rather than instantaneous across future steps; hence a residual lag remains between local observation and global belief update. Nevertheless, the measured gains indicate that the coordination overhead is justified under the tested regimes.

6 Ablation Study

We investigated the contribution on the core designs of our agent system by disabling each of them and test on each configuration for 10 runs and take the mean value for comparison. One is the communication mechanism, we added a `CommunicationBlackhole` that absorbs all outgoing messages and replaced the message receiving function with an empty function, to remove communication only while keeping other parts of the system intact. The other core design is each of our Agent's local policy and planner, we implemented a `MinimalAgent` that only has the basic mobility and survival functions provided by the `Group7AgentBase`, with a simple greedy policy to pick up the nearest tile and fill the nearest hole, without any of the additional logic in the original design.

Then we ran the ablation experiments with the settings and the results are shown in the table below.

Design Ablation	Parameter 1	Parameter 2	Parameter 3
Full System	**1086.5**	**1397.2**	**2440.8**
No Communication	402.2	371.7	61.4
BaseAgent Only	459.5	748.5	545.6

Table 2: Ablation Experiment Results (Mean Rewards over 10 runs)

We observe three consistent patterns. First, removing communication causes a major degradation in all settings: relative to the full system, mean reward drops by 63.0% (Set 1), 73.4% (Set 2), and 97.5% (Set 3). This confirms that under partial observability, local policy quality alone cannot compensate for missing shared situational awareness. Also the catastrophic drop in Parameter 3 is caused by a high probability of agents dying from out-of-fuel, which is a direct consequence of missing our initial zone search coordinated by communication.

Second, replacing specialized agents with `AgentMinimal`-style base behavior also reduces performance substantially: 57.7%, 46.4%, and 77.6% drops for Sets 1–3 respectively. Hence, the common infrastructure in `Group7AgentBase` is necessary but not sufficient; specialized planners and memory models contribute significantly beyond baseline survivability.

7 Conclusion

This project developed a six-agent Tileworld system with a shared engineering backbone (`Group7AgentBase`) and agent-specific decision modules. The implemented framework unifies path planning, fuel-safe control, and a structured communication protocol (G7P2), while allowing each agent to introduce distinct planning and memory mechanisms.

Empirically, our results show that the complete system consistently outperforms ablated variants across all tested parameter sets. The ablation study further demonstrates that communication and policy specialization are complementary: removing either substantially harms final reward.

The current implementation still has limitations. Richer planners / exploration strategies can introduce nontrivial runtime overhead on large maps, and some systems rely heavily on SS message which have a high bandwidth cost. Future work should therefore target
(i) automated dynamic strategy orchestration that takes computation and communication bandwidth into account,
(ii) adaptive communication budgets that manages bandwidth-urgency trade-offs, and
(iii) tighter integration between global coordination signals and local plan repair to improve robustness under extreme dynamism.

References

[1] Jie Zhang. Agent decision making. Lecture slides for AI6125: Multi-Agent Systems, 2026. NTU AI6125 Lecture Slides.

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Appendix

[AgentHanny]
Deriviation of the object survival probability estimation and the online MLE process for hazard estimation.

$$p_{alive} = e^{-\lambda \Delta t} \quad \in (0, 1]$$

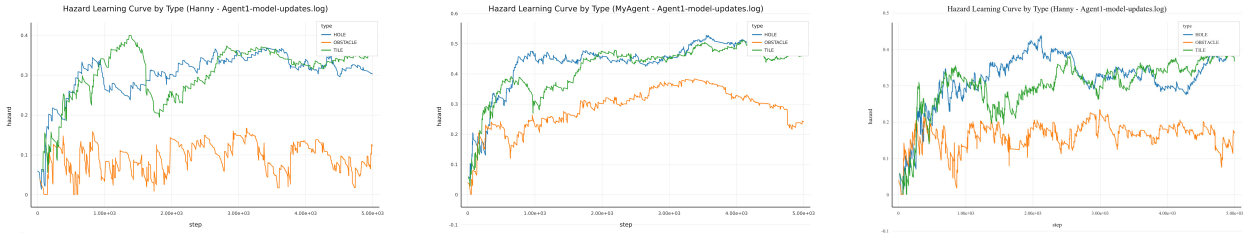
Here, Δt is the estimated time of arrival from the current time stamp, and λ is a reflecting environment $hazard$ that affects the agent's expectation of an object still alive. This $harzard$ will be different across different experiment setups, such as world size and object creation distribution.

We model each observe on an object as a Bernoulli trial, with matching observation with possibility $P_{match} = \exp(-\lambda \Delta t)$ and a mismatch with possibility $P_{mismatch} = 1 - \exp(-\lambda \Delta t)$. Then we can use the maximum likelihood estimation (MLE) to find the best λ that fits the observation data. By computing the gradient of the negative log-likelihood $l(P)$, we can then perform the MLE via mini-batch stochastic gradient decent (SGD) with update rule as follows:

$$\text{Upon Match} \quad \frac{\partial \ell}{\partial \lambda} = -\Delta t \quad \Longrightarrow \quad \lambda \leftarrow \max\left(0, \lambda - \eta \Delta t\right)$$

$$\text{Upon MisMatch} \quad \frac{\partial \ell}{\partial \lambda} = \frac{\Delta t \, e^{-\lambda \Delta t}}{1 - e^{-\lambda \Delta t}} = \frac{\Delta t}{e^{\lambda \Delta t} - 1} \quad \Longrightarrow \quad \lambda \leftarrow \lambda + \eta \cdot \frac{\Delta t}{e^{\lambda \Delta t} - 1 + \epsilon}$$

By using learning rate schedule and gradient ceiling and clipping, the lambda is then converged to a stable range after 1000 steps on online training, as the curves shown in the figure below.



(a) λ vs Steps - Parameters 1 (b) λ vs Steps - Parameters 2 (c) λ vs Steps - Parameters 3

Figure 2: Hazard Figure Across Different Environment Settings (without smoothing).

[AgentHanny]
Expected utility deriviation on a plan with multiple targets:

$$EU_{plan} = \sum_{pickup} p_{tile-alive} \cdot 0.20 + \sum_{fill} p_{hole-alive} \cdot 1.00 - 0.07 \cdot Distance - FuelPenalty$$